# Yukun Du


Current city: Athens, GA
Email: yukdu@outlook.com
Website: https://benniedu.github.io/

Last Updated: October 23, 2026

## Employment

Aug 2024 - Current — Limited-Term Assistant Professor
*Department of Mathematics*, **University of Georgia**, Athens, GA
Research Mentor: Sa'ar Hersonsky

## Education

Sep 2019 - Jun 2024 — Ph.D. in Mathematics
*Department of Mathematics*, **University of California, Davis**, Davis, CA
Advisor: Michael Kapovich

Sep 2016 - Jul 2019 — M.S. in Mathematics
*Department of Mathematical Science*, **Tsinghua University**, Beijing
Advisor: Runliang Lin

Sep 2012 - Jul 2016 — B.S. in Mathematics & Applied Mathematics
*Department of Mathematical Science*, **Tsinghua University**, Beijing
Advisor: Runliang Lin

## Research Interests

- Geometric Group Theory
- Hyperbolic Geometry
- Lie Groups and Symmetric Spaces
- Trees and Tits Buildings

## Publications

- In preparation:

  [1] Y. Du. *Lattices of Minimal Covolume in* $\mathrm{PSL}_2$ *over Dyadic Local Fields*. In preparation. 2026.

- Preprints:

  [2] Y. Du. *Two-Generator Discrete Subgroups of Tree Automorphism Groups*. Submitted for publication. 2026. arXiv: 2606.06824.
  [3] Y. Du, F. Hao, and K. Zhu. *Profinite Non-Rigidity of Arithmetic Lattices and the Kähler Property*. Preprint. 2026. arXiv: 2609.31523.
  [4] Y. Du and S. Hersonsky. *Terminal Geometry of Rank-Two Nielsen Reduction on Simplicial Trees*. Submitted for publication. 2025. arXiv: 2512.00957.
  [5] Y. Du. *Busemann-Selberg Functions and Completeness for Dirichlet-Selberg domains in* $\mathrm{SL}(n,\mathbb{R})/\mathrm{SO}(n)$. Submitted for publication. 2024. arXiv: 2412.06987.

- Published:

  [6] Y. Du. "Geometry of Selberg's bisectors in the symmetric space $SL(n,\mathbb{R})/SO(n,\mathbb{R})$". In: *J. Lond. Math. Soc.* 110.4 (Oct. 2024).
  [7] R. Lin and Y. Du. "Generalized Darboux Transformation for the Discrete Kadomtsev–Petviashvili Equation with Self-Consistent Sources". In: *Theoretical and Mathematical Physics* 196 (2018).

## Code Repositories

- Compute Dirichlet-Selberg Domains from generators in the group $\mathrm{SL}(3,\mathbb{R})$.
  GitHub: https://github.com/BennieDu/Compute-Dirichlet-Selberg-Domains

## Selected Talks at Conferences & Seminars

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| Sep 2026 | Geometric Group Theory Seminars at the Ohio State University<br>Columbus, OH |
| Jul 2026 | Regensburg GAP Days 2<br>Regensburg, Germany |
| Jun 2026 | GAGTA 2026: Group Actions in Dynamics<br>New York City, NY |
| Oct 2025 | AMS Sectional Meeting: Special Session on Topology and Group Actions<br>St. Louis, MO |
| Jun 2025 | Midwest Summer School in Geometry, Topology, and Dynamics<br>Madison, WI |
| Feb 2025 | Geometry and Topology Seminar at Georgia Institute of Technology<br>Atlanta, GA |
| Oct 2023 | Geometry and Topology seminar at Yale University<br>New Haven, CT |
| May 2023 | Graduate Student Geometry and Topology Conference<br>Philadelphia, PA |
| Oct 2020 | Moduli Spaces Seminar at the University of Melbourne<br>Zoom |

## Honors

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| 2018 | Hua Luo-Geng Scholarship |
| 2017 | Second-Class Scholarship |
| 2015 | Science & Technology Innovation Scholarship |
| 2012 | Freshmen Scholarship |

## Teaching Experience

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| Aug 2024 - Current | Instructor of Record<br>*Department of Mathematics*, **University of Georgia**, Athens, GA<br>Taught Precalculus (MATH 1113, Fall 2024 and Fall 2025), Calculus I (MATH 2250, Spring 2026), Calculus II (MATH 2260, Fall 2026), Calculus III (MATH 2270, Spring 2025), and Complex Variables (MATH 4150, Fall 2025). |
| Aug - Sep 2021,<br>Aug - Sep 2023 | Summer Quarter Instructor<br>*Department of Mathematics*, **University of California, Davis**, Davis, CA<br>Taught Calculus (MAT 021A) and BioCalculus (MAT 017C). |
| Mar 2020 - Jun 2024 | Teaching Assistant<br>*Department of Mathematics*, **University of California, Davis**, Davis, CA<br>TA for courses including Precalculus (MAT 012), Short Calculus (MAT 016 series), Calculus for BioSci (MAT 017 series), and Modern Algebra (MAT 150A). |
| Sep 2016 - Jan 2019 | Teaching Assistant<br>*Department of Mathematical Science*, **Tsinghua University**, Beijing<br>TA for courses including Calculus, Linear Algebra, and Complex Functions. |

## Advising

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| Spring 2026 | Directed Reading Group Mentor to John Calame (Undergraduate Student, UGA)<br>Topic: Octonions and Cayley Integers |
| Spring 2025 | Directed Reading Group Mentor to Nicolas Trullols Sandino (Undergraduate Student, UGA)<br>Topic: Hyperbolic Geometry and Spacetime |
| 2021 - 2022 | Directed Reading Group Mentor to Tiching Kao (Undergraduate Student, UC Davis)<br>Topic: Groups and Symmetry |

## Outreach and Public Service

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| Apr 2026 | Event Volunteer, 2nd Sonia Kovalevsky Day<br>*Department of Mathematics*, **University of Georgia**<br>Contributed to the organization and facilitation of a day-long mathematics outreach event aimed at inspiring local middle school students to explore STEM disciplines. |

## Skills and Abilities

- **Mathematics**: Mathematica, SageMath, R
- **Programming**: C++, Python
- **Documentation**: LaTeX, Microsoft Office

## Languages

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| English | CEFR C1 (TOEFL iBT 102) | ●●●●○ |
| Chinese (Mandarin) | Native | ●●●●● |
| Japanese | CEFR B2 (JLPT N2 149/180) | ●●●○○ |

## Interests

- Running/Hiking
- Drawing
- Comics/Anime